

Case Study 3 – AeroParts Manufacturing Inc. (Manufacturing Sector)

1. Company Overview & Mission Statement

Company Name: AeroParts Manufacturing Inc.

Mission: "To design and manufacture world-class aerospace components that ensure safety, efficiency, and innovation in air travel."

Company Summary:

AeroParts Manufacturing Inc. is a mid-sized manufacturer specializing in precision aerospace components for commercial and defense contractors. The organization integrates advanced design, engineering, and industrial control technologies to deliver high-tolerance parts to global clients. AeroParts operates under aerospace industry regulations (e.g., AS9100) and has a growing need for cybersecurity controls across IT and OT environments.

2. Organizational Structure

Employees: ~600

Departments:

- Design & Engineering
- Manufacturing Operations
- Procurement & Supply Chain
- Quality Assurance & Compliance
- IT & OT Support
- HR & Corporate Services
- Executive Leadership

Third-Party Relationships:

- Contract engineers and CAD design contractors
- Raw materials suppliers (global)
- OEM aerospace manufacturers (clients)
- Vendor portal operators for supply chain logistics

3. Facilities and Operations

Physical Sites:

- 2 manufacturing plants (Plant #1: primary production, Plant #2: precision machining)
- 1 engineering/design center (adjacent to HQ)

Key Operations:

- CAD/CAM design of aerospace components
- Multi-stage precision fabrication and QA
- ERP-managed logistics and inventory
- Supplier and vendor collaboration via portal

Security & Compliance Frameworks:

- AS9100D, ISO 27001, ITAR (select contracts)
- Internal audit cycles every 12 months

4. IT Infrastructure

Data Center:

- Primary data center at HQ with hardened physical access, monitored server cages

Servers:

- 8 design/CAD servers for 3D modeling and simulation
- 6 ERP servers managing inventory, scheduling, and finance
- 4 vendor-facing servers for portal access and documentation

Cloud Services:

- Limited cloud presence (vendor portal, document collaboration)
- Data classification policy restricts cloud storage for design files

Endpoints (~600):

- Workstations and laptops across departments
- Industrial control terminals (IoT) across production floors

Network & Remote Access:

- 1 VPN gateway for vendor and supplier access (role-limited)
- Segmented VLANs for OT (IoT), IT (corporate), and Guest networks

Security Devices:

- 3 perimeter firewalls (HQ + plants)
- Internal VLAN firewalls
- IDS/IPS inline at datacenter ingress
- Centralized EDR platform, USB device monitoring in progress

5. Mission-Critical Systems

System	Criticality	Description
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CAD/CAM Design Systems	Critical	Engineering of components, essential to production lifecycle
ERP System	High	Inventory, production scheduling, financials
IoT Manufacturing Control	High	Direct control of machinery and robotics on floor
Vendor Portal	Medium	External supplier access and documentation exchange

6. Data Classification & Access Controls

Sensitive Data Types:

- Proprietary CAD designs and blueprints
- Client contract data (including ITAR-controlled data)
- Vendor contractual and financial data

Access Control Methods:

- Role-Based Access Control (RBAC) with least privilege enforcement
- MFA for all remote access
- Periodic USB audits and device whitelisting
- Contractor access through sandboxed VLAN with time-based accounts

User Profiles:

- Engineers: Full CAD system access, restricted network browsing
- Operators: IoT terminal access, ERP production workflows
- Vendors: Portal-only access via VPN with MFA
- IT/OT Admins: Elevated access, dual-authentication enforced

7. Recent Threat Events

- Suspected industrial espionage involving targeted scans of CAD repositories
- USB-delivered malware incident via contract engineer workstation
- Inadequate segmentation between IoT and corporate network led to lateral movement
- Vendor access portal experienced a 3-hour outage during a maintenance error

8. Downtime Tolerances & Business Impact

System	Max Tolerable Downtime	RTO	RPO	Key Impacts
Design Systems	48 hours	24 hr	2 hr	Client project delays, design iteration freeze
ERP System	48 hours	24 hr	4 hr	Material shortages, missed production targets
IoT Control Systems	24 hours	8 hr	1 hr	Production line halt, shipment and SLA breaches
Vendor Portal	72 hours	48 hr	12 hr	Delayed communication, resupply issues

9. Incident Scenario (August 18, 2025)

At 7:50 AM, two production lines at Plant #2 shut down after IoT control terminals lost connectivity. Alerts showed malformed packets flooding the control VLAN. IT traced the traffic to a design center workstation infected via unauthorized USB use.

Incident Findings:

- Malware scanned IoT endpoints for ERP integration vulnerabilities
- Malware likely originated from a contract engineer's flash drive
- Weak authentication protocols on IoT devices were exploited

Impact:

- 6-hour downtime, \$250,000 in delayed production
- No customer data compromised, but design systems were quarantined for analysis

Response:

- Isolated affected VLANs, removed malware from endpoints
- Initiated ISO 27001 incident reporting and forensic audit
- Accelerated USB control rollout and segmentation overhaul

10. Additional Considerations for Students

Students may consider:

- Improving segmentation between IT and OT environments
- Creating USB usage policy and enforcement strategies
- Evaluating the trust model for contractor access
- Mapping risks to operational continuity and compliance (e.g., AS9100)
- Developing layered incident response plans for IT vs IoT systems